GODEMENT RESOLUTION OF A SHEAF OF MODULES BY FLASQUE MODULES WITH STALKS A HOMOTOPY EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof refers to a result 'mentioned in the introduction to this section'; the sentence is reproduced as written and the result in question is a navigation pointer carried as a stub, while the reduction that makes it applicable is inside the counted span.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-godement-resolution is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Complex associated to a cosimplicial object (Dold-Kan, cosimplicial form). Simplicial, Section simplicial-section-dold-kan-cosimplicial, invoked at proof lines 7045-7046. Supplies the identification of the displayed complex with the one attached to the cosimplicial object f_*f^* applied iteratively to F.
- f^* is exact and exactness can be tested after pulling back. Modules, Lemma modules-lemma-abelian, invoked at proof lines 7054-7060. Supplies that a complex of O_X-modules is exact if and only if its pullback is, which is what makes the stalkwise argument work.
- Homotopy equivalence of a constant cosimplicial object with its associated cosimplicial object. Simplicial, results on the standard simplicial homotopy named in the introduction to the section and used at proof lines 7061-7070. Supplies the stalkwise homotopy equivalence statement of the lemma.

Source reference. Original lemma 01AG: sources/stacks-chow/tags.txt, source lines 1689–1689. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0FKK: sources/stacks-chow/tags.txt, source lines 19465–19465. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original section 019H: sources/stacks-chow/tags.txt, source lines 1652–1652. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. For every sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ there is a resolution*

$$0 \to \mathcal{F} \to f_*f^*\mathcal{F} \to f_*f^*f_*f^*\mathcal{F} \to f_*f^*f_*f^*f_*f^*\mathcal{F} \to \ldots$$

functorial in $\mathcal{F}$ such that each term $f_*f^* \ldots f_*f^*\mathcal{F}$ is a flasque $\mathcal{O}_X$-module and such that for all $x \in X$ the map

$$\mathcal{F}_x[0] \to \Big((f_*f^*\mathcal{F})_x \to (f_*f^*f_*f^*\mathcal{F})_x \to (f_*f^*f_*f^*f_*f^*\mathcal{F})_x \to \ldots\Big)$$

is a homotopy equivalence in the category of complexes of $\mathcal{O}_{X,x}$-modules.

Proof. The complex $f_*f^*\mathcal{F} \to f_*f^*f_*f^*\mathcal{F} \to f_*f^*f_*f^*f_*f^*\mathcal{F} \to \ldots$ is the complex associated to the cosimplicial object with terms $f_*f^*\mathcal{F}, f_*f^*f_*f^*\mathcal{F}, f_*f^*f_*f^*f_*f^*\mathcal{F}, \ldots$ described above, see Simplicial, Section 019H. The augmentation gives rise to the map $\mathcal{F} \to f_*f^*\mathcal{F}$ as indicated. For any abelian sheaf $\mathcal{H}$ on X_{disc} the pushforward $f_*\mathcal{H}$ is flasque because X_{disc} is a discrete space and the pushforward of a flasque sheaf is flasque. Hence the terms of the complex are flasque $\mathcal{O}_X$-modules.

If $x \in X_{disc} = X$ is a point, then $(f^*\mathcal{G})_x = \mathcal{G}_x$ for any $\mathcal{O}_X$-module $\mathcal{G}$. Hence f^* is an exact functor and a complex of $\mathcal{O}_X$-modules $\mathcal{G}_1 \to \mathcal{G}_2 \to \mathcal{G}_3$ is exact if and only if $f^*\mathcal{G}_1 \to f^*\mathcal{G}_2 \to f^*\mathcal{G}_3$ is exact (see Modules, Lemma 01AG). The result mentioned in the introduction to this section proves the pullback by f^* gives a homotopy equivalence from the constant cosimplicial object $f^*\mathcal{F}$ to the cosimplicial object with terms $f_*f^*\mathcal{F}, f_*f^*f_*f^*\mathcal{F}, f_*f^*f_*f^*f_*f^*\mathcal{F}, \ldots$. By Simplicial, Lemma 0FKK we obtain that

$$f^*\mathcal{F}[0] \to \Big(f^*f_*f^*\mathcal{F} \to f^*f_*f^*f_*f^*\mathcal{F} \to f^*f_*f^*f_*f^*f_*f^*\mathcal{F} \to \ldots\Big)$$

is a homotopy equivalence. This immediately implies the two remaining statements of the lemma. □